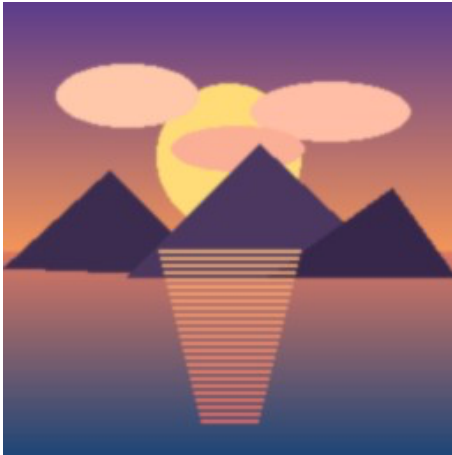


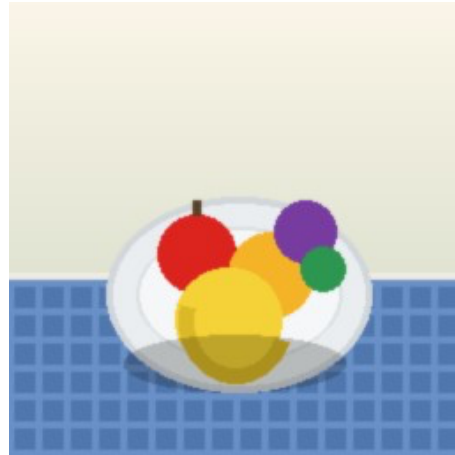
Assignment 4 Report

K-Means Clustering for Image Compression

This report implements K-Means clustering from scratch and applies it to color quantization on two simple colored images. Each pixel is treated as a three-dimensional RGB data point. The algorithm is tested for $K = 2, 3, 5, 10, 15$, and 20 .



Original Image 1: Sunset



Original Image 2: Fruit

Why these images were selected: they are easy to understand, colorful, quick to process, and still show a clear visual improvement as K increases.

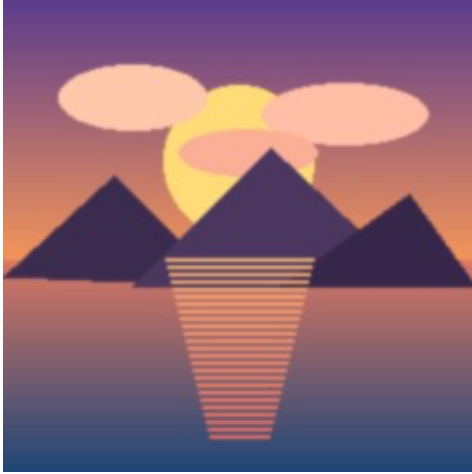
Method summary: initialize centroids, assign each pixel to the nearest centroid, recompute the centroids as cluster means, and repeat until convergence. The final compressed image is created by replacing each pixel with the RGB centroid of its assigned cluster.

Part 1: $K = 2$ Results and Final Centroids

Image	Cluster	R	G	B
Sunset	1	94	73	111
Sunset	2	209	141	120
Fruit	1	236	233	211
Fruit	2	107	125	162

K = 2 Image Comparison

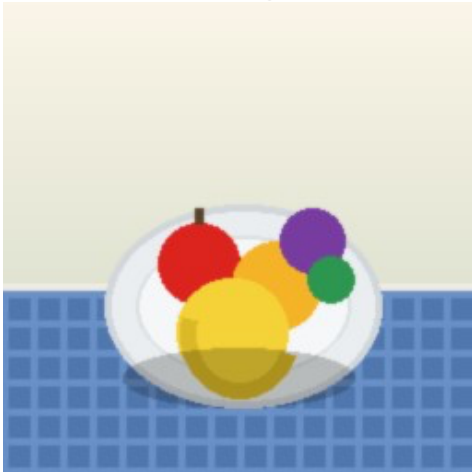
Sunset: Original



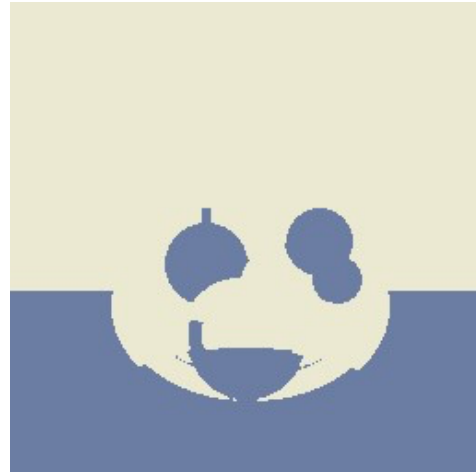
Sunset: K=2



Fruit: Original



Fruit: K=2



With $K = 2$, each image is reduced to only two colors. The major regions are still visible, but many details disappear. This is why $K = 2$ gives the strongest compression but the weakest visual fidelity.

Part 2: Analysis

Is the compressed image similar to the original image? Yes, but only in a rough sense when $K = 2$. The compressed image keeps the large shapes and main color groups, but small color transitions and fine details are lost.

Visual appearance comment: In the sunset image, $K = 2$ separates the scene into broad dark and light regions, so the sky and mountains remain recognizable. In the fruit image, $K = 2$ keeps the table and wall contrast, but the individual fruit colors are merged too aggressively.

Image 1: Sunset Results for All K Values

The contact sheet below includes the original image and the compressed results for K = 2, 3, 5, 10, 15, and 20.



Original



K=2



K=3



K=5



K=10



K=15



K=20

K	Iterations	MSE	Observation
2	7	754.86	Very strong compression; many details lost.
3	19	388.64	Main shapes are clearer; colors still simplified.
5	4	165.65	Main shapes are clearer; colors still simplified.
10	15	57.40	Good balance between compression and quality.
15	24	29.01	Very close to original with richer colors.
20	21	20.71	Very close to original with richer colors.

Image 2: Fruit Results for All K Values

The contact sheet below includes the original image and the compressed results for K = 2, 3, 5, 10, 15, and 20.



Original



K=2



K=3



K=5



K=10



K=15

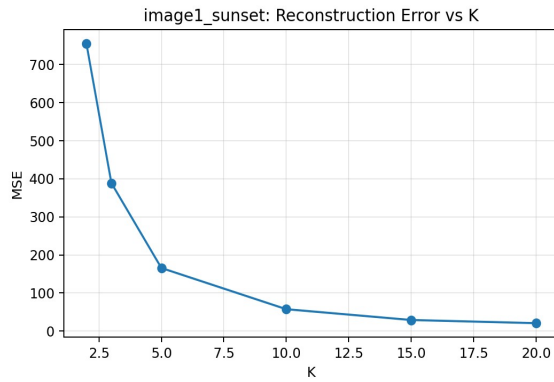


K=20

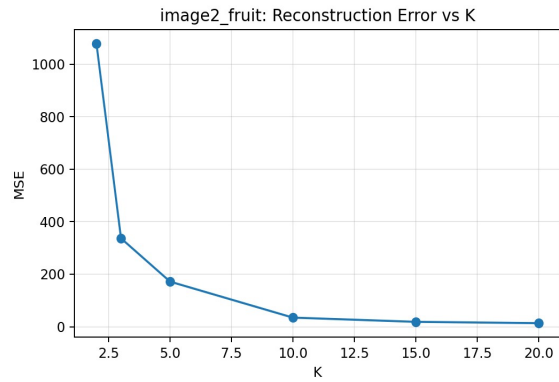
K	Iterations	MSE	Observation
2	8	1078.22	Very strong compression; many details lost.
3	3	336.71	Main shapes are clearer; colors still simplified.
5	3	171.76	Main shapes are clearer; colors still simplified.
10	9	34.71	Good balance between compression and quality.
15	9	18.67	Very close to original with richer colors.
20	6	13.59	Very close to original with richer colors.

Part 3: Effect of Varying K

As K increases, the compressed images preserve more color information. This produces smoother gradients, clearer boundaries, and stronger visual similarity to the original image. The MSE graphs below support the visual observation: reconstruction error falls steadily as K increases.



Sunset: MSE decreases as K increases.



Fruit: MSE decreases as K increases.

Comment on the results for each value of K:

- K = 2: only the strongest color separation remains; image content is recognizable but heavily simplified.
- K = 3: one extra centroid helps separate another region, so the image structure improves.
- K = 5: more object regions and color bands become visible.
- K = 10: a strong improvement; most important details and transitions are preserved.
- K = 15: the result is very close to the original for both images.
- K = 20: the closest result among the tested values, with the richest color representation.

Does the compressed image improve as K increases?

Yes. In both test images, increasing K improves visual quality and lowers reconstruction error. For the sunset image, MSE falls from 754.86 at K=2 to 20.71 at K=20. For the fruit image, MSE falls from 1078.22 to 13.59.

Effect of increasing K on the three required points:

Aspect	Effect of increasing K
Visual quality	Improves because the image can represent more shades and boundaries.
Similarity to original image	Increases because more centroid colors better approximate the original pixels.
Color representation	Becomes richer and more faithful because the palette contains more distinct RGB centroids.

Conclusion

K-Means is effective for image compression through color quantization. A small K gives stronger compression but poorer quality, while a larger K preserves more detail and improves similarity to the original image. Among the tested values, K = 15 and K = 20 gave the best visual results.